Lab 9: Socratic Tutoring with LLMs: Improving an Algorithm

Introduction to the Task

In this activity, you will use a Large Language Model (LLM) such as **ChatGPT** or **Copilot** to support your design of a decision tree algorithm for classifying customer credit card prediction. The LLM will act as a ***Socratic tutor*** — posing questions that guide your thinking, challenge your assumptions, and help you construct your experiment step-by-step.

You will interact with the LLM across five structured stages. At each stage, you will provide a prompt and receive questions from the LLM. You can ask follow-up questions, but only if they prompt the LLM to ask you further clarifying or challenging questions. You **must not** ask the LLM to give you direct answers or write the algorithm for you.

Your goal is to evaluate how far the questions posed by the LLM help you in designing your experiment. You may keep notes or reflections on which questions were most helpful and why.

1. Stage 1: Understanding the Problem

Prompt to LLM:

" *I have trained a logistic regression model to predict whether **a passenger will or will not survive on the titanic**. I want to improve the results . Can you ask me three questions to help me think of ways to improve the results? Make your answers relatively short.* "

2. Stage 2: Decomposing the Algorithm

Prompt to LLM:

" *Help me break down the different solutions to improve the experiment. Ask me three questions on algorithms or other solutions that can improve the results. Make your answers relatively short.*"

3. **Stage 3: Designing the Algorithm**

Prompt to LLM:

" I want to restart the experiment in scikit learn. Can you ask me three questions on what I need to do? The questions should help me make a good design of the algorithm? Make your answers relatively short. "

4. **Stage 4: Evaluating the Design**

Prompt to LLM:

" *Let's evaluate my algorithm. Ask me how I would test the improved performance and what metrics I should use? Ask question that will help me chose the right metrics. Make your answers relatively short "*